

2. Some facts from analysis. Optimality conditions

Optimization T2026

Master in Fundamental Principles of Data Science



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Scalar product

Let $\mathbf{x} = (x_1, \dots, x_n)^T$ and $\mathbf{y} = (y_1, \dots, y_n)^T$ belong to $\mathbb{R}^n$.

Scalar (or dot) product

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$$

Euclidean norm

$$\| \mathbf{x} \| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{x_1^2 + \dots + x_n^2}$$

Euclidean distance

$$d(\mathbf{x}, \mathbf{y}) = \| \mathbf{x} - \mathbf{y} \| = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Cosine of the angle

$$\cos \theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\| \mathbf{x} \| \| \mathbf{y} \|}, \quad \mathbf{x}, \mathbf{y} \neq \mathbf{0}$$

Orthogonality

$$\mathbf{x} \perp \mathbf{y} \quad \Leftrightarrow \quad \mathbf{x} \cdot \mathbf{y} = 0$$

Other useful distances include Minkowski, city-block and correlation distances.

Lines in $\mathbb{R}^n$

A **line** in $\mathbb{R}^n$ is determined by a point $\mathbf{a} = (a_1, \dots, a_n)^T \in \mathbb{R}^n$ and the direction $\mathbf{v} = (v_1, \dots, v_n)^T \in \mathbb{R}^n$. It is equal to the set of points $\mathbf{x} = (x_1, \dots, x_n)^T \in \mathbb{R}^n$ that satisfy

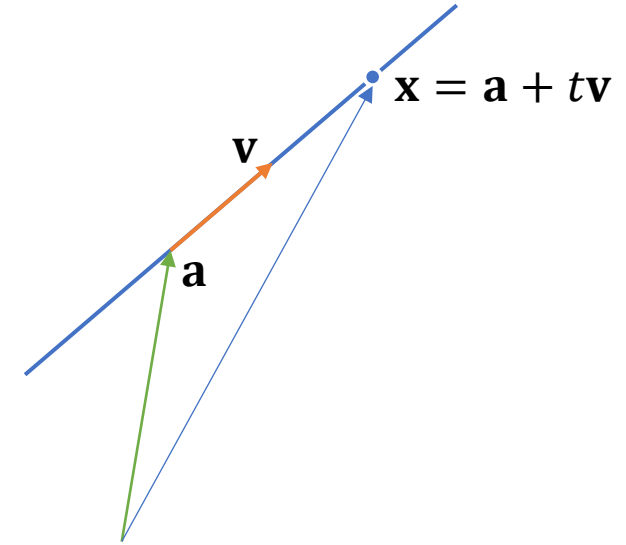
$$\mathbf{x} = \mathbf{a} + t\mathbf{v}, \quad t \in \mathbb{R} \quad \Leftrightarrow \quad \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} + t \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$$

Equivalently, the line can be written as

$$\frac{x_1 - a_1}{v_1} = \dots = \frac{x_n - a_n}{v_n}$$

In $\mathbb{R}^2$, the line is given by the linear equation

$$A_1x_1 + A_2x_2 + A_0 = 0, \quad \text{where } A_1 = v_2, A_2 = -v_1, A_0 = -a_1v_2 + a_2v_1$$



Hyperplanes in $\mathbb{R}^n$

A **hyperplane** is determined by a point $\mathbf{a} = (a_1, \dots, a_n)^T$ and $(n - 1)$ vectors in $\mathbb{R}^n$

$$\mathbf{u}^1 = (u_1^1, \dots, u_n^1)^T, \dots, \mathbf{u}^{n-1} = (u_1^{n-1}, \dots, u_n^{n-1})^T,$$

and is equal to the set of points $\mathbf{x} = (x_1, \dots, x_n)^T$ of the form

$$\mathbf{x} = \mathbf{a} + t_1 \mathbf{u}^1 + \dots + t_{n-1} \mathbf{u}^{n-1}, t_i \in \mathbb{R} \Leftrightarrow \begin{pmatrix} x_1 \\ \dots \\ x_n \end{pmatrix} = \begin{pmatrix} a_1 \\ \dots \\ a_n \end{pmatrix} + t_1 \begin{pmatrix} u_1^1 \\ \dots \\ u_n^1 \end{pmatrix} + \dots + t_{n-1} \begin{pmatrix} u_1^{n-1} \\ \dots \\ u_n^{n-1} \end{pmatrix}$$

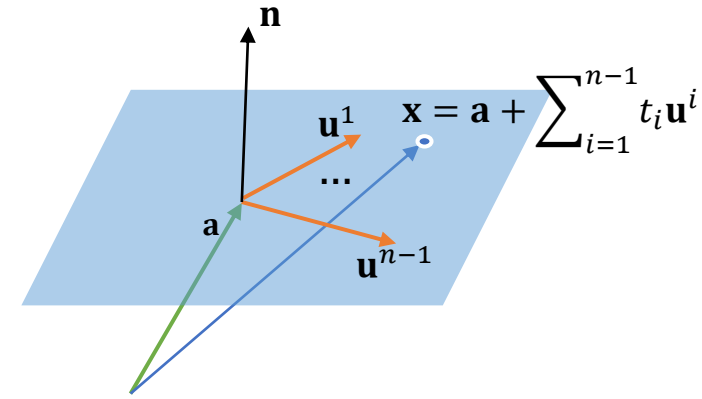
The equation above can be written as

$$\begin{vmatrix} x_1 - a_1 & \dots & x_n - a_n \\ u_1^1 & \dots & u_n^1 \\ \dots & \dots & \dots \\ u_1^{n-1} & \dots & u_n^{n-1} \end{vmatrix} = 0$$

or as

$$A_0 + A_1 x_1 + \dots + A_n x_n = 0$$

where $\mathbf{n} = (A_1, \dots, A_n)^T$ is the **normal vector** to the plane, i.e., $(\mathbf{x} - \mathbf{a})^T \mathbf{n} = 0$.



Real-valued functions: continuity

Consider a **real-valued function**

$$f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$$

- The **domain** $D := D(f)$ of f is the set of points in $\mathbb{R}^n$ where the function is defined
- The **graph of f** is defined as

$$\mathcal{G}(f) := \{(\mathbf{x}^T, w)^T \in \mathbb{R}^{n+1} : \mathbf{x} = (x_1, \dots, x_n)^T \in D, w = f(\mathbf{x}) \in \mathbb{R}\}$$

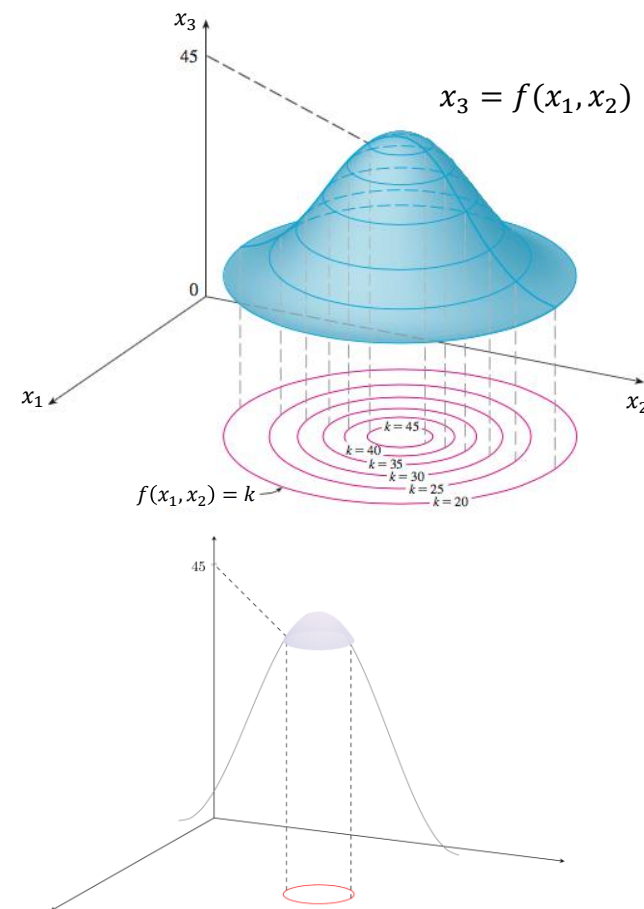
- The **level set of f (of level $c \in \mathbb{R}$)** is defined as

$$L_c = \{\mathbf{x} \in D : f(\mathbf{x}) = c\} \subset \mathbb{R}^n$$

The function f is **continuous** at $\mathbf{a} \in D$ if and only if

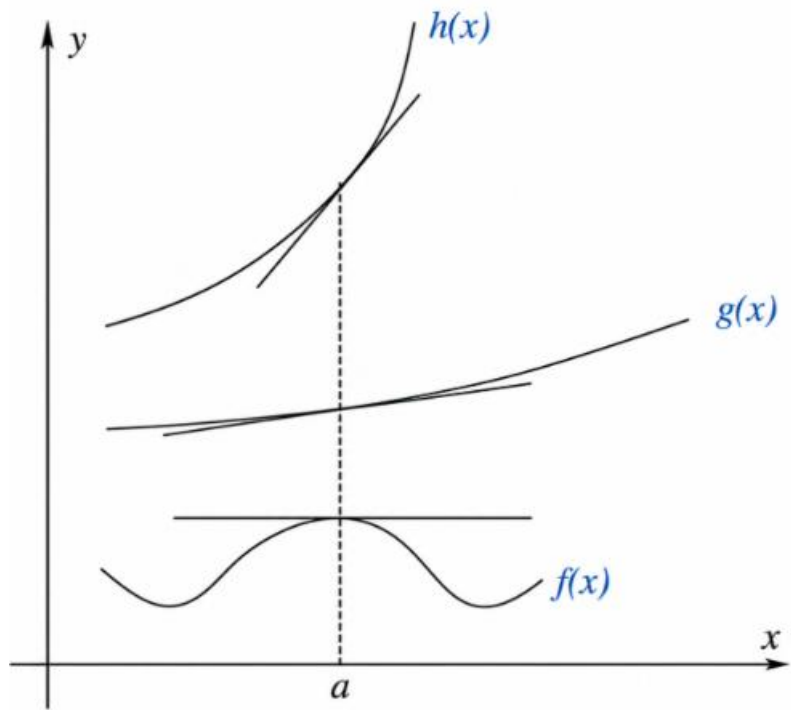
$$\lim_{D \ni \mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = f(\mathbf{a})$$

(whenever the limit makes sense)



Derivative in dimension $n = 1$

The **derivative** of $y = f(x)$ measures the instantaneous rate of change of y with respect to x .



Geometrically

Secant lines approach the tangent line.

The derivative is the slope of the tangent line.

Analytically

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

f is **differentiable** at $a \in D(f)$ if the derivative $f'(a)$ exists.

Derivative in dimension $n = 1$

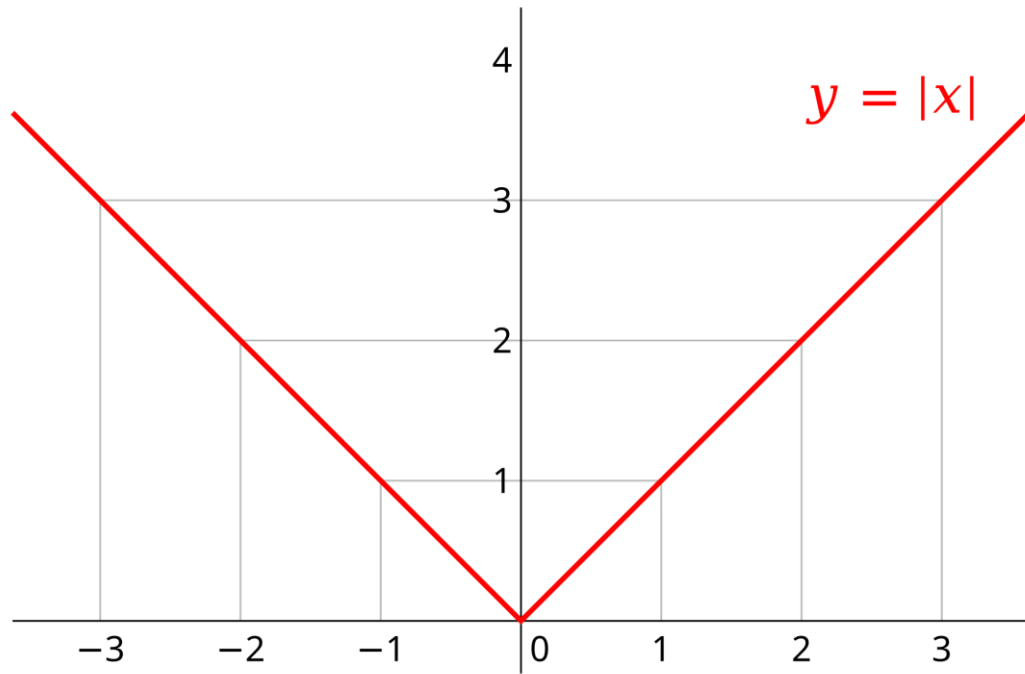
The definition $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$ of the derivative gives:

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a) - f'(a)h}{h} = 0$$

The tangent line gives the **best linear approximation** to f near a :

$$f(a+h) \approx f(a) + f'(a)h \quad (h \text{ small})$$

Continuity and differentiability in dimension $n = 1$



Theorem. If f is differentiable at a , then f is continuous at a .

The converse need not hold.

For example, $f(x) = |x|$ is continuous at 0 but is not differentiable there.

A function $f: (b, c) \rightarrow \mathbb{R}$ is **differentiable on (b, c)** if it is differentiable at every point of the interval.

Differentiability in $\mathbb{R}^n$, $n > 1$: partial derivatives

Let $f: D \subset \mathbb{R}^n \rightarrow \mathbb{R}$, $f(\mathbf{x}) = f(x_1, \dots, x_n)$, and let $\mathbf{a} = (a_1, \dots, a_n)^T \in D$ be an interior point.

The **partial derivative** with respect to x_j measures the change in f when x_j varies and the other coordinates are fixed. Formally:

$$f_{x_j}(\mathbf{a}) = \frac{\partial f}{\partial x_j}(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_j + h, \dots, a_n) - f(a_1, \dots, a_j, \dots, a_n)}{h}$$

The partial derivative is defined whenever this limit exists.

Differentiability in $\mathbb{R}^n$, $n \geq 1$: gradient vector

Let f and $\mathbf{a}$ be as above. The vector formed by the partial derivatives of f at the point $\mathbf{a}$ is the **gradient vector**

$$\nabla f(\mathbf{a}) = \left(\frac{\partial f}{\partial x_1}(\mathbf{a}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{a}) \right)^T$$

We say that f is **differentiable at $\mathbf{a}$** if all partial derivatives of f exist at $\mathbf{a}$ and

$$\lim_{\mathbf{h} \rightarrow 0} \frac{f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) - \nabla f(\mathbf{a})^T \mathbf{h}}{\|\mathbf{h}\|} = 0, \quad \mathbf{h} \in \mathbb{R}^n$$

One can show that if the partial derivatives of f are continuous in a neighborhood of $\mathbf{a}$, then f is differentiable at $\mathbf{a}$

From the definition it follows that the **best linear approximation** of the differentiable function f at a point $\mathbf{a} \in D$ is given by the linear function

$$f(\mathbf{x} + \mathbf{a}) \approx \mathcal{L}(\mathbf{x}) = f(\mathbf{a}) + \nabla f(\mathbf{a})^T (\mathbf{x} - \mathbf{a})$$

Gradient vector: linear approximation

Example: Let $f(\mathbf{x}) = f(x, y, z) = ze^x\sqrt{y}$. Estimate the value of $f(0.01, 24.8, 1.02)$.

We take $\mathbf{a} = (0, 25, 1)^T$ and we use the linear approximation of f given by

$$\begin{aligned} L(\mathbf{x}) &= f(\mathbf{a}) + (\nabla f(\mathbf{a}))^T (\mathbf{x} - \mathbf{a}) = 5 + (5, 1/10, 5) \begin{pmatrix} x - 0 \\ y - 25 \\ z - 1 \end{pmatrix} \\ &= 5 + 5x + \frac{1}{10}(y - 25) + 5(z - 1) \end{aligned}$$

Finally

$$L(0.01, 24.8, 1.02) = 5.13 \approx f(0.01, 24.8, 1.02) = 5.1306.$$

Differentiability in $\mathbb{R}^n$, $n \geq 1$: directional derivative

Partial derivatives measure changes along the coordinate axes. A **directional derivative** measures the change along any chosen direction.

Choose a **unit vector** $\mathbf{v} = (v_1, \dots, v_n)^T$, so that $\|\mathbf{v}\| = 1$.

The directional derivative of f at $\mathbf{a}$ in the direction $\mathbf{v}$ is

$$D_{\mathbf{v}}f(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{v}) - f(\mathbf{a})}{h}$$

The derivative is defined whenever this limit exists.

Differentiability in $\mathbb{R}^n$, $n \geq 1$: directional derivative (cont.)

Chain rule. Let $\alpha: (a, b) \rightarrow \mathbb{R}^n$ be a differentiable curve and let f be differentiable along it

$$\alpha(t) = (\alpha_1(t), \dots, \alpha_n(t))^T, \quad f(\alpha(t)) = f(\alpha_1(t), \dots, \alpha_n(t))$$

Then:

$$\frac{d}{dt} f(\alpha(t)) = \sum_{j=1}^n \frac{\partial f}{\partial x_j}(\alpha(t)) \alpha_j'(t)$$

Corollary. Apply the chain rule to the line $\mathbf{x} = \mathbf{a} + t\mathbf{v}$:

$$D_{\mathbf{v}} f(\mathbf{a}) = \left. \frac{d}{dt} f(\mathbf{a} + t\mathbf{v}) \right|_{t=0} = \nabla f(\mathbf{a})^T \mathbf{v}$$

The gradient vector

Theorem. Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function at $\mathbf{a} \in D$, and $\mathbf{u} \in \mathbb{R}^n$ is a unitary vector. The following statements hold.

(a)

$$D_{\mathbf{u}}f(\mathbf{a}) = (\nabla f(\mathbf{a}))^T \cdot \mathbf{u} = \|\nabla f(\mathbf{a})\| \cos \theta,$$

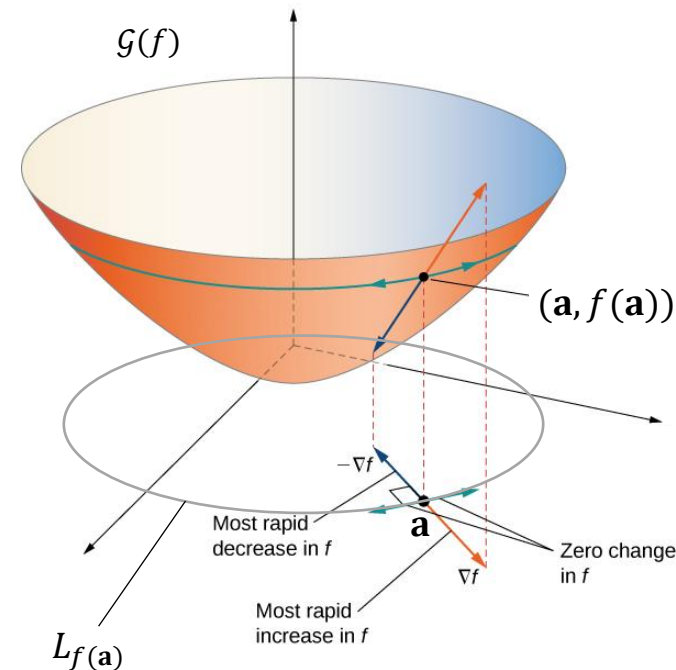
where θ is the angle between $\mathbf{u}$ and $\nabla f(\mathbf{a})$.

(b) The gradient vector $\nabla f(\mathbf{a})$ gives the maximum direction variation of f at the point $\mathbf{a}$.

(c) The gradient vector at the point $\mathbf{a} \in D$ is orthogonal to the level curve passing through $\mathbf{a}$.

Proof. Statements (a) and (b) are direct. Let $\mathbf{r}(t) = (x_1(t), x_2(t), \dots, x_n(t))^T$ be the level curve passing through $\mathbf{a} \in D$. Then

$$0 = \frac{d}{dt}f(\mathbf{r}(t)) = \dots = \nabla f(\mathbf{r}(t))^T \mathbf{r}'(t).$$



Real-valued functions: higher-order derivatives

Let $f: D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a function, and let $\mathbf{a} = (a_1, \dots, a_n)^T$ be an interior point of D .

Assume f and its first partial derivatives are differentiable at $\mathbf{a}$.

The **second-order partial derivatives** are defined by

$$f_{x_i x_j}(\mathbf{a}) = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a}) := \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_j} \right) (\mathbf{a}), \quad i, j = 1, \dots, n$$

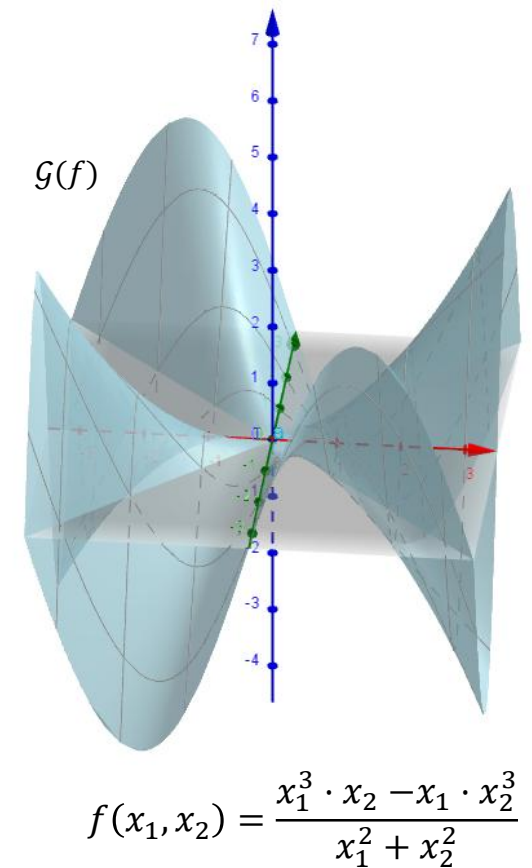
Real-valued functions: the Hessian

The **Hessian matrix** at $\mathbf{a} \in D$ is

$$\nabla^2 f(\mathbf{a}) = \begin{pmatrix} f_{x_1 x_1}(\mathbf{a}) & \cdots & f_{x_1 x_n}(\mathbf{a}) \\ \vdots & \ddots & \vdots \\ f_{x_n x_1}(\mathbf{a}) & \cdots & f_{x_n x_n}(\mathbf{a}) \end{pmatrix}$$

Schwarz's theorem. If D is open and $f \in C^2(D)$, then the Hessian is symmetric:

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{a})$$



Taylor's expansion for f : linear part

Assume all needed regularity of f in the next slides. When we say that

$$L(\mathbf{x}) = f(\mathbf{a}) + (\nabla f(\mathbf{a}))^T (\mathbf{x} - \mathbf{a})$$

is the linear approximation of the function f at a point $\mathbf{a}$ we mean (roughly speaking) that

$$f(\mathbf{x}) = f(\mathbf{a}) + (\nabla f(\mathbf{a}))^T (\mathbf{x} - \mathbf{a}) + \mathcal{O}(|\mathbf{x} - \mathbf{a}|^2).$$

This is known as Taylor's expansion of order one.

Taylor's expansion for f : quadratic part

The second order approximation of f at the point $\mathbf{a} \in D$ is

$$f(\mathbf{x}) = f(\mathbf{a}) + (\nabla f(\mathbf{a}))^T (\mathbf{x} - \mathbf{a}) + \frac{1}{2} \nabla^2(f)(\mathbf{a})(\mathbf{x} - \mathbf{a})^2 + \mathcal{O}(|\mathbf{x} - \mathbf{a}|^3)$$

where the value of $\nabla^2(f)(\mathbf{a})(\mathbf{x} - \mathbf{a})^2 \in \mathbb{R}$ is given by

$$(x_1 - a_1, \dots, x_n - a_n) \begin{pmatrix} \frac{\partial^2 f(\mathbf{a})}{\partial x_1^2} & \cdots & \frac{\partial^2 f(\mathbf{a})}{\partial x_n \partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\mathbf{a})}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 f(\mathbf{a})}{\partial x_n^2} \end{pmatrix} \begin{pmatrix} x_1 - a_1 \\ \vdots \\ x_n - a_n \end{pmatrix}.$$

We say that

$$Q(\mathbf{x}) = f(\mathbf{a}) + (\nabla f(\mathbf{a}))^T (\mathbf{x} - \mathbf{a}) + \frac{1}{2} \nabla^2(f)(\mathbf{a})(\mathbf{x} - \mathbf{a})^2$$

is the **quadratic approximation** of the function f at a point $\mathbf{a}$.

Quadratic functions

As it will be clear during the course it is crucial for getting the **optimality conditions** to determine whether the expression $\nabla^2(f)(\mathbf{a})(\mathbf{x} - \mathbf{a})^2$ is positive or negative for $\mathbf{x} \in \mathbb{R}^n$ (or at least for $\mathbf{x}$ near $\mathbf{a}$). Remember that $\nabla^2(f)(\mathbf{a})$ is a symmetric matrix. The key notion is the following.

Definition. Let Q any symmetric $n \times n$ matrix.

- ① Q is **positive semidefinite** (PSD) if and only if $\mathbf{x}^T Q \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$.
- ② Q is **positive definite** (PD) if and only if $\mathbf{x}^T Q \mathbf{x} > 0$ for all $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x} \neq 0$.

Theorem. Let Q be a symmetric $n \times n$ matrix. Then all the eigenvalues of Q are real. Moreover, Q is PD if and only if all the eigenvalues of Q are positive.

Local and global extrema

Let $D \subset \mathbb{R}^n$ be open, and let $f: D \rightarrow \mathbb{R}$.

$$B(\mathbf{x}^*, \varepsilon) = \{\mathbf{x} \in \mathbb{R}^n: \|\mathbf{x} - \mathbf{x}^*\| \leq \varepsilon\}$$

Local minimum. The point $\mathbf{x}^* \in D$ is a local minimizer ($f(\mathbf{x}^*)$ local minimum) if some $\varepsilon > 0$ satisfies $B(\mathbf{x}^*, \varepsilon) \subset D$ and

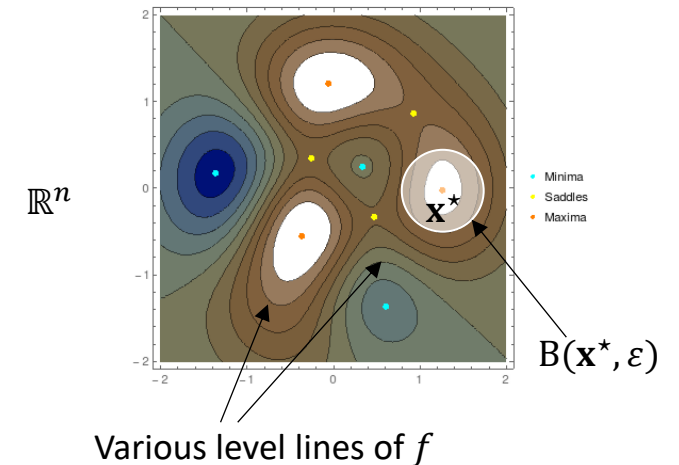
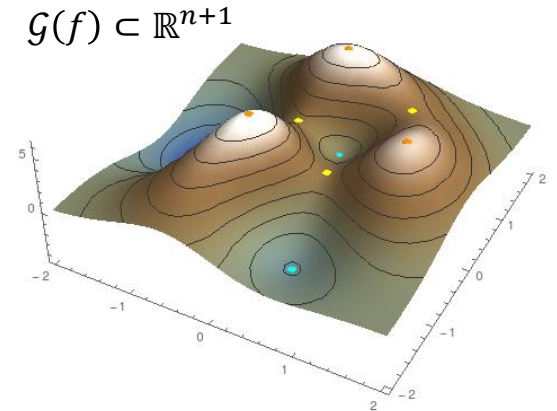
$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for all } \mathbf{x} \in B(\mathbf{x}^*, \varepsilon)$$

Global minimum. The point $\mathbf{x}^* \in D$ is a global minimizer ($f(\mathbf{x}^*)$ global minimum) if

$$f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for all } \mathbf{x} \in D$$

The minimum is **strict** when the inequality is strict for $\mathbf{x} \neq \mathbf{x}^*$.

For **maxima**, reverse the inequalities. Minima and maxima are called **extrema**.



Local and global extrema: existence

Theorem (Weierstrass). Let

$$f : K \subset \mathbb{R}^n \longrightarrow \mathbb{R},$$

be continuous function such that K is **compact** (closed and bounded), then f is bounded (there exist M such that $|f(x)| < M$ for all $x \in K$) and f attains its maximum and minimum values on K .

A necessary condition for minima

Theorem. Let $D \subset \mathbb{R}^n$ be an open domain and let $f : D \rightarrow \mathbb{R}$ be a $\mathcal{C}^1$ function (i.e., continuously differentiable). Assume $\mathbf{x}^* \in D$ is a local minimum for f . Then

$$\nabla f(\mathbf{x}^*) = 0.$$

Proof. We argue by contradiction. By hypothesis, if $\lambda > 0$ is small enough, for all $\mathbf{v} \in \mathbb{R}^n$, we have that

$$f(\mathbf{x}^*) \leq f(\mathbf{x}^* + \lambda \mathbf{v}). \quad (1)$$

Fix $\mathbf{v}$ (w.l.o.g. $\|\mathbf{v}\| = 1$). Define $F_{\mathbf{v}}(\lambda) := F(\lambda) = f(\mathbf{x}^* + \lambda \mathbf{v})$. Then, (1) writes as

$$F(0) \leq F(\lambda), \quad \forall |\lambda| < \delta.$$

for some $\delta > 0$. From the Mean Value Theorem, we have

$$F(\lambda) = F(0) + F'(\theta\lambda)\lambda,$$

for some $\theta \in [0, 1]$.

Proof (continue).

If $F'(0) > 0$, then, since F' is a continuous function, we have that, if $|\lambda|$ is small enough, $F'(\theta\lambda) > 0$ for all $\theta \in [0, 1]$. Hence, we can find a $\lambda < 0$ small enough such that

$$F(\lambda) = F(0) + F'(\theta\lambda)\lambda < F(0),$$

a contradiction. Arguing similarly with $F'(0) < 0$ we conclude $F'(0) = 0$, or equivalently,

$$F'(0) = (\nabla f(\mathbf{x}^*))^T \mathbf{v} = 0.$$

Since $\mathbf{v}$ is an arbitrary unitary vector, we must have:

$$\nabla f(\mathbf{x}^*) = 0.$$

Remark. Similar arguments show that if $\mathbf{x}^* \in D$ is a local maxima of a $\mathcal{C}^1$ map f , then

$$\nabla f(\mathbf{x}^*) = 0.$$

Sufficient conditions for minima

Theorem. Let $D \subset \mathbb{R}^n$ open and let $f : D \rightarrow \mathbb{R}$ be a $\mathcal{C}^2$ map (twice continuously differentiable). Assume

$$\nabla f(\mathbf{x}^*) = 0, \quad \mathbf{z}^T \nabla^2(f)(\mathbf{x}^*) \mathbf{z} > 0, \quad \forall \mathbf{z} \neq 0.$$

Then f has a **strict** local minimum at $\mathbf{x}^*$.

Exercise. Prove the Theorem above (Use the second order Taylor expansion of f around $\mathbf{x}^*$).

Exercise. Prove that the converse of Theorem above is not true. Consider the family of one-dimensional real valued functions $f_p(x) = x^p$, $p \geq 1$.

Necessary and sufficient conditions for minima

Theorem Let $D \subset \mathbb{R}^n$ open and let $f : D \rightarrow \mathbb{R}$ be a $\mathcal{C}^2$ map (twice continuously differentiable). The following statements hold

- (a) (Necessary conditions) Assume $\mathbf{x}^* \in D$ is a local minima of f .
Then

$$\nabla f(\mathbf{x}^*) = 0 \quad \text{and} \quad \mathbf{z}^T \nabla^2(f)(\mathbf{x}^*) \mathbf{z} \geq 0, \quad \forall \mathbf{z} \in \mathbb{R}^n.$$

- (b) (Sufficient conditions) Assume $\nabla f(\mathbf{x}^*) = 0$. Assume also that there exists $\varepsilon > 0$ such that for all $\mathbf{x} \in \mathbf{B}(\mathbf{x}^*, \varepsilon)$ we have

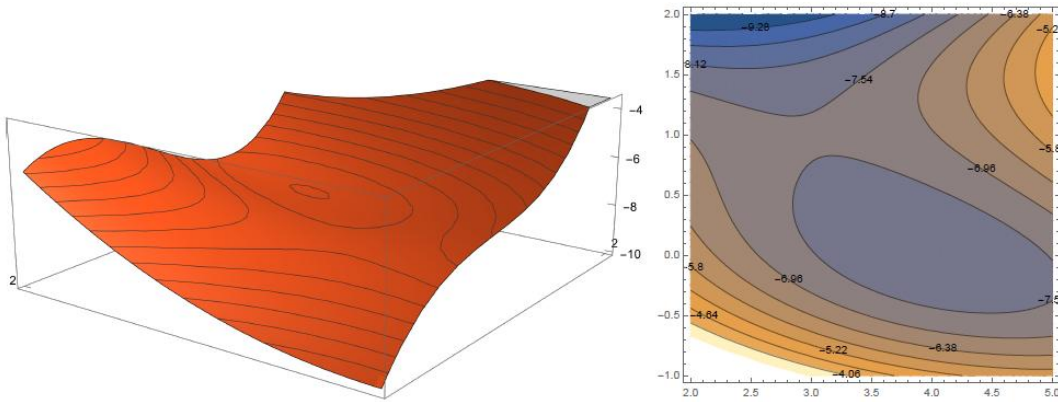
$$\mathbf{z}^T \nabla^2(f)(\mathbf{x}) \mathbf{z} \geq 0, \quad \forall \mathbf{z} \in \mathbb{R}^n.$$

Then, $\mathbf{x}^* \in D$ is a local minima of f .

An example

Find the (local) extrema for the polynomial function

$$f(x, y) = \frac{1}{2}x^2 + xy + 2y^2 - 4x - 4y - y^3.$$



Computations. We have that

$$\nabla f(x, y) = (x + y - 4, x + 4y - 4 - 3y^2)^T \quad \text{and} \quad \nabla^2(f)(x, y) = \begin{pmatrix} 1 & 1 \\ 1 & 4 - 6y \end{pmatrix}.$$

Easily

$$\nabla f(x, y) = 0 \iff \mathbf{x}_1 = (4, 0)^T \text{ and } \mathbf{x}_2 = (3, 1)^T,$$

and

$$\nabla^2 f(\mathbf{x}_1) = \begin{pmatrix} 1 & 1 \\ 1 & 4 \end{pmatrix}, \quad \nabla^2 f(\mathbf{x}_2) = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}.$$

$\nabla^2 f(\mathbf{x}_1)$ is positive definite, $\nabla^2 f(\mathbf{x}_2)$ is indefinite.

Conclusion. The only extrema is the local minimum is $\mathbf{x}_1 = (4, 0)^T$.

Problem (problems must be delivered by the end of the semester)

$$f_{\lambda}(x_1, x_2) = (x_2 - x_1^2)^2 + x_1^6 - \lambda x_1^4, \quad \lambda \in \mathbb{R}$$

Justify all conclusions analytically.

(a) Show that $(0, 0)$ is stationary for every λ . Compute its Hessian and eigenvalues.

What does the second-order test tell you?

(b) Find and classify all stationary points for each λ . Justify the classification directly whenever the Hessian test is inconclusive.

(c) Prove that the function tends to $+\infty$ as $x_1^2 + x_2^2 \rightarrow \infty$. Find all global minimizers and the minimum value. Describe what changes as λ passes through 0.

(d) For each nonzero $\mathbf{v} = (a_1, b_1)^T$, restrict the function to the line $(x_1, x_2)^T = (ta, tb)^T$. For which λ is $t = 0$ a strict local minimum for every $\mathbf{v}$? Compare with part **(b)**.

(e) When the conclusions differ, find a smooth curve γ with $\gamma(0) = (0, 0)^T$ such that the value of the function along the curve are smaller than at the origin for all sufficiently small $t \neq 0$. Explain why the neighbourhood in the line test may depend on the direction.